THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a triangle is 180 cm². If the altitude on the side of 24 cm is 15 cm, the altitude on the side of 30 cm (in cm) is: 1

(a) 12 **(b)** 15 **(c)** 14 **(d)** 6

2. A rhombus has perimeter 148 cm and one diagonal 24 cm. Its area is: 1

(a) 420 cm² **(b)** 888 cm² **(c)** 840 cm² **(d)** 1680 cm²

3. A 64 m by 40 m rectangular lawn has a path 5 m wide inside along its boundary. The area of the lawn left is: 1

(a) 2460 m² **(b)** 2065 m² **(c)** 1520 m² **(d)** 1620 m²

4. A triangle has sides 39 cm, 80 cm and 89 cm. Its area is: 1

(a) 208 cm² **(b)** 1735 cm² **(c)** 1560 cm² **(d)** 3120 cm²

5. The perimeter of a rectangle is 150 cm and the length is 4 times the breadth. Its area is: 1

(a) 900 cm² **(b)** 1406 cm² **(c)** 915 cm² **(d)** 870 cm²

6. An isosceles trapezium has parallel sides 14 cm and 30 cm and non-parallel sides 17 cm. Its area is: 1

(a) 330 cm² **(b)** 660 cm² **(c)** 340 cm² **(d)** 374 cm²

7. The length of a rectangle is increased by 10% and its breadth decreased by 20%. If the original area is 1500 cm², the new area (in cm²) is: 1

(a) 1500 **(b)** 1350 **(c)** 1650 **(d)** 1320

8. The area of a triangle is 210 cm². If the altitude on the side of 28 cm is 15 cm, the altitude on the side of 35 cm (in cm) is: 1

(a) 14 **(b)** 6 **(c)** 12 **(d)** 15

9. **Assertion (A):** A trapezium with parallel sides 8 cm and 12 cm and height 5 cm has area 50 cm^2. 1
Reason (R): Area of a trapezium = (sum of parallel sides) × height.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Two triangles with equal bases and equal heights must be congruent. 1
Reason (R): Triangles with equal bases and equal heights have equal areas.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A rectangle's length is increased by 40% and breadth decreased by 25%. By what percent does the area change? (original area 100 units) — 2

12. The area of a triangle is 210 cm² and one side is 30 cm. Find the altitude on that side and the area of a triangle with the same base and double the altitude. — 2

13. The perimeter of a rectangle is 160 cm and its length exceeds its breadth by 24 cm. Find its area. — 2

14. A rectangle's length is increased by 20% and breadth decreased by 30%. By what percent does the area change? (original area 100 units) — 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The perimeter of a rectangular plot is 180 m and its length is 5 times its breadth. Find its area and the cost of fencing at ₹ 20 per metre. — 3

16. An isosceles trapezium has parallel sides 16 cm and 30 cm and each slant side 25 cm. Find its height and area. — 3

17. A garden 60 m by 40 m has a rectangular pond 15 m by 10 m inside it and a path 2 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. — 3

OR

The perimeter of a rectangular plot is 210 m and its length is 4 times its breadth. Find its area and the cost of fencing at ₹ 10 per metre. — 3

18. An isosceles trapezium has parallel sides 18 cm and 34 cm and each slant side 17 cm. Find its height and area. — 3

OR

A garden 36 m by 28 m has a rectangular pond 9 m by 6 m inside it and a path 1 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A parallelogram-shaped lawn in a park has a base of 24 m and a height of 15 m. Grass planting costs ₹40 per m^2. — 5

(i) Find the area of the lawn.
(ii) Find the area of a triangular flower bed with the same base and height.
(iii) Another parallelogram has the same area and a base of 18 m; find its height.
(iv) Find the cost of planting grass on the lawn and on half of it.

20. The area of a rhombus is 24 cm² and one diagonal is 6 cm. Find the other diagonal, the side, and the altitude. — 5

OR

A trapezium ABCD has parallel sides AB = 12 cm and DC = 28 cm and height 9 cm. A parallelogram of base 12 cm and the same height is cut off from it, leaving a triangle. Find the area of the trapezium, the parallelogram and the remaining triangle. — 5

END OF QUESTION PAPER — Total: 40 Marks